ROGERS MUKIIBI

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Personal Profile

Electrical Engineering graduate from Makerere University with applied experience across AI/ML systems, IoT, and electrical distribution networks. Published co-author on an IEEE Access paper on an IoT-enabled multimodal honeybee monitoring system. Built and deployed AI systems including a RAG-based chatbot and machine learning models for solar irradiance estimation, alongside hardware work spanning PCB design to distribution-network modeling in DIgSILENT PowerFactory. Proficient in Python, C++, SQL, and modern data science/ML tooling. Passionate about applying AI and Quantum computing to solve real-world engineering and sustainability challenges.

Education

Makerere University — **Kampala, Uganda**
BSc Electrical Engineering — *January 2022 – Feb 2026*

WorldQuant University — **Online**
Data Science Certificate — *December 2023 - January 2024*

Crested Secondary School — **Kampala, Uganda**
UACE - Physics, Chemistry, Mathematics (AAA) — *February 2019 – April 2021*

Work Experience

Associate R & D Power Systems Engineer.
Auri Energy — *July 2025 – present*

- Developed and validated power system models to simulate distribution network behavior under varying load and fault conditions.
- Analyzed power quality metrics including voltage sags/swells, harmonic distortion (up to 40th order), and voltage unbalance in line with IEC standards.
- Contributed to the design and implementation of SCADA-based monitoring systems for real-time grid visibility and performance tracking.
- Supported digital twin development of distribution networks, integrating simulation insights with operational data for improved system reliability.

Experience

AI Software Development Intern
Xeno Investments — *Jul 2025 - Aug 2025*

- Built a RAG-based chatbot to handle customer queries on Xeno services, integrating ChromaDB vector search with LLM response generation.
- Implemented intent classification to route simple queries efficiently, reducing unnecessary LLM usage.
- Designed a flexible embedding pipeline supporting both Google (*gemini-embedding-001*) and local (*BAAI/bge-m3*) models.
- Developed session-based memory and performance logging (latency tracking) to improve response quality and system reliability.

Research Assistant
Marconi Machine Learning Laboratory — *April 2024 - Aug 2025*

- Developed and deployed AI-based models to estimate solar irradiation for Uganda, directly addressing the 3-20% positive bias in existing satellite-derived data (CAMS RAD, NASA Power) that introduces significant financial and operational risks in PV project planning.
- Collaborating with GIZ and the Ministry of Energy to develop and disseminate a free and open-source solar irradiation dataset, specifically fine-tuned for Uganda to enhance regional solar energy planning and investment decisions.

Undergraduate Intern
Internet of Things - Research & Applications Laboratory (IoT-RA) — *Nov 2023 – March 2024*

- Designed and fabricated a **2-layer PCB** for a smart beehive monitoring system, optimizing a wire-based prototype into a robust, field-ready design that reduced assembly time by 83% (from 2 hours to 20 minutes).
- Integrated an ADXL345 accelerometer on the beehive sensor unit controlled by a Raspberry Pi to capture bee vibrations and monitor bee behaviour.
- Developed signal processing pipeline implementing **Fast Fourier Transform** (FFT) analysis of beehive vibrations, achieving real-time data acquisition at 10kHz sampling rate.
- Co-authored resulting research as **"A Smart IoT-enabled Multimodal System for Honeybee Keepers in Uganda,"** published in **IEEE Access** (Early Access, 2026).
- A more detailed portfolio of my work at the lab is available at IOTra Portfolio link.

Undergraduate Intern

netLabs!UG *Jun 2023 – Aug 2023*

- Designed and implemented **Proxmox-based server clusters** from electronic waste, extending hardware lifespan and promoting sustainable IT practices.
- Deployed **Docker containerized application** for resource-intensive engineering software (e.g., Blender), providing cloud-based access for students lacking sufficient local computational power.
- Awarded Second Place at the Makerere Engineering Society Open Day for innovative solution leveraging e-waste into robust server infrastructure, enhancing access to critical design software for the student body.

Publications

2026 A. Turyagyenda, A. Katumba, R. Akol, M. Nsabagwa, M. E. Mkiramweni, and **R. Mukiibi**, *"A Smart IoT-enabled Multimodal System for Honeybee Keepers in Uganda,"* **IEEE Access**, doi: 10.1109/ACCESS.2026.3709045 (Early Access).

2025 **R. Mukiibi**, *"Developing AI Models for Estimating Solar Irradiation for Improved Solar Energy Planning in Uganda,"* B.Sc. Dissertation, Makerere University, Kampala. View dissertation.

Skills

Power Systems & Grid Engineering
Analysis & Simulation: DIgSILENT PowerFactory, MATLAB/Simulink, Power System Modeling
Grid Studies: Load Flow, Voltage Profiles, Harmonic Analysis (up to 40th order), Voltage Unbalance (VUF)
Focus Areas: Distribution Systems, SCADA Integration, Grid Reliability, Power Quality
Software & Simulation Engineering
Programming: Python, Rust, C/C++
Modeling: Numerical Simulation, System Modeling, Digital Twin Concepts
Data & Signal Processing: FFT, Time-Series Analysis, Power Quality Metrics
AI Systems: Domain-Specific Chatbots, Knowledge-Based AI Systems
Embedded & Hardware Systems
Hardware Design: PCB Design, Circuit Design, Hardware Prototyping
Tools: KiCad, Altium, LTSpice, Multisim, Quartus (AHDL), Lattice Radiant (VHDL)
Microcontrollers: Arduino, Raspberry Pi, LoRa (Pycom)
Systems Integration: Sensor Interfacing, Data Acquisition Systems
Engineering Tools & Platforms
Design: AutoCAD Electrical
Workflow: Technical Documentation, System Specification, Engineering Collaboration

Professional Memberships

IEEE Member Member of the Institute of Electrical and Electronics Engineers (IEEE), including the IEEE Power & Energy Society and Industrial Electronics Society.

Makerere Engineering Society Actively involved in events like Makerere Engineering Society Open Day.

Volunteering

2024	**Project Proposal Reviewer**	*EPICS in IEEE*
2024	**Webmaster AEE Makerere Chapter**	*Association of Energy Engineers*
2023	**Student Ambassador IEEE IES Region 8**	*IEEE Industrial Electronics Society*
2022	**Secretary IEEE CAS Makerere Chapter**	*IEEE Circuits and Systems Young Professionals*

Awards

2025	**DataCamp Data Scientist Certification**	DS0026002460117
2024	**Makerere Engineering Society Open Day**	First Runners Up
2022	**IEEE Xtreme Coding Competition**	Ranked 1st in Uganda